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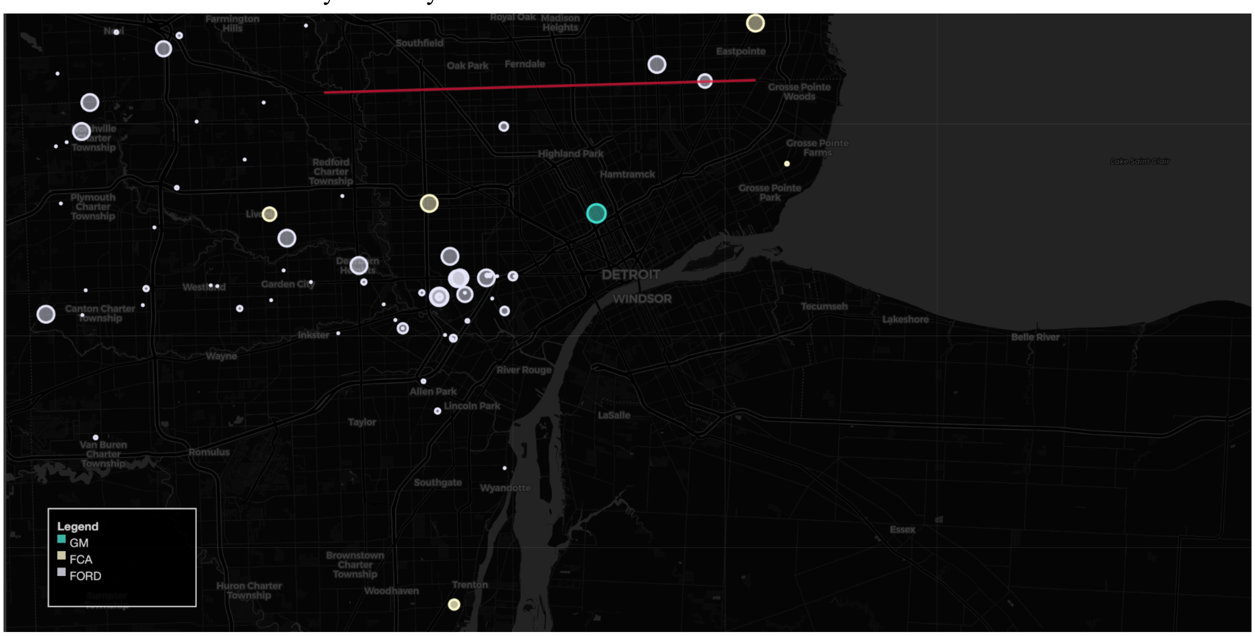
MoTown Mo' Buses!!!

Mapping Auto Industry Jobs in Detroit

To better understand the spatial relationship between employment and access, a map was created to visualize the locations of major automobile companies operating within Detroit. Given the central role that Ford, GM, and FCA continue to play in the city's economy, business-level data was filtered to display only sites belonging to these "Detroit Three" manufacturers. Each site was plotted using circle markers, where the size of each marker corresponds to the number of local employees and the color represents the company.

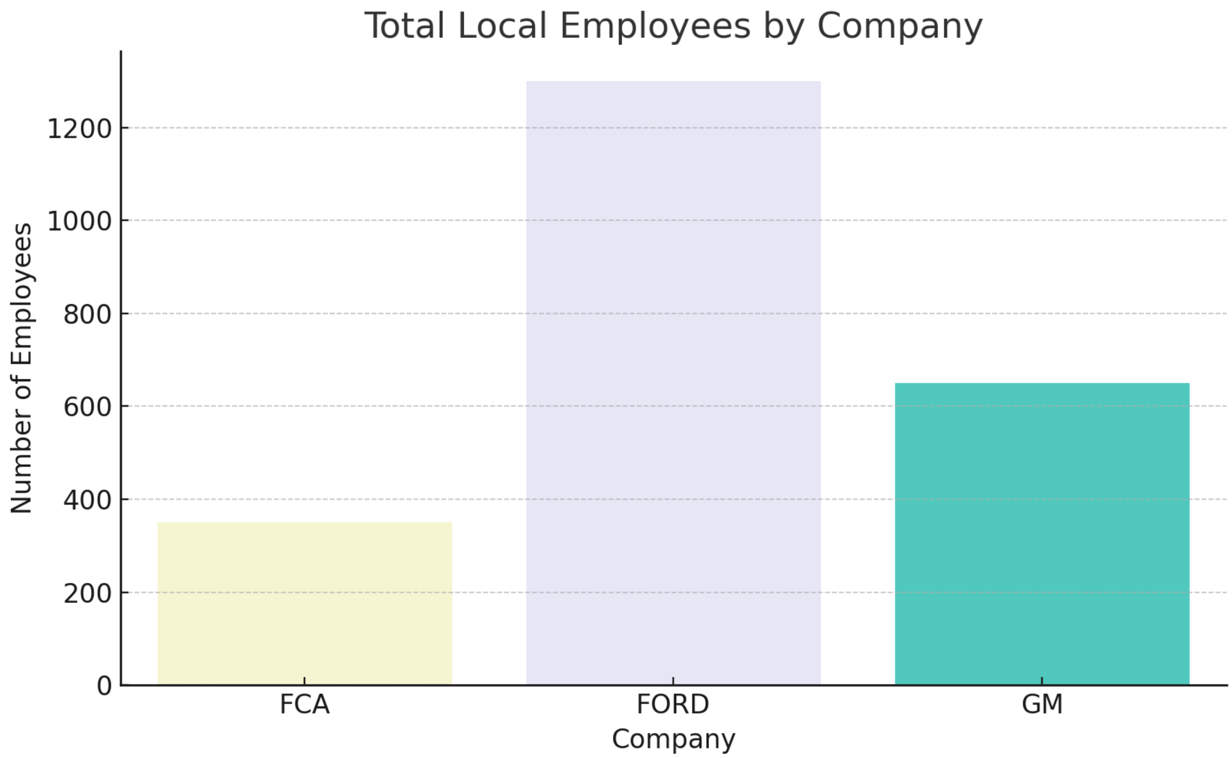
This visualization offers a clear picture of how concentrated and unevenly distributed these employment centers are. While many sites remain within Detroit's boundaries, a number are located closer to the suburban edges or in areas with limited public transit coverage. For transit-dependent residents, particularly those without access to a personal vehicle, these job sites may be physically present but practically out of reach. The result is a visible gap between where jobs exist and where the city's most transit-reliant populations live.

The map reinforces broader findings in this study related to spatial mismatch and transportation inequality. Despite the presence of major employers, structural barriers such as unreliable transit and historic segregation continue to limit access to opportunity. Without significant changes in transportation infrastructure or policies aimed at bridging these divides, Detroiters who rely on public transit will remain disconnected from key sources of employment even in industries that define the city's identity.



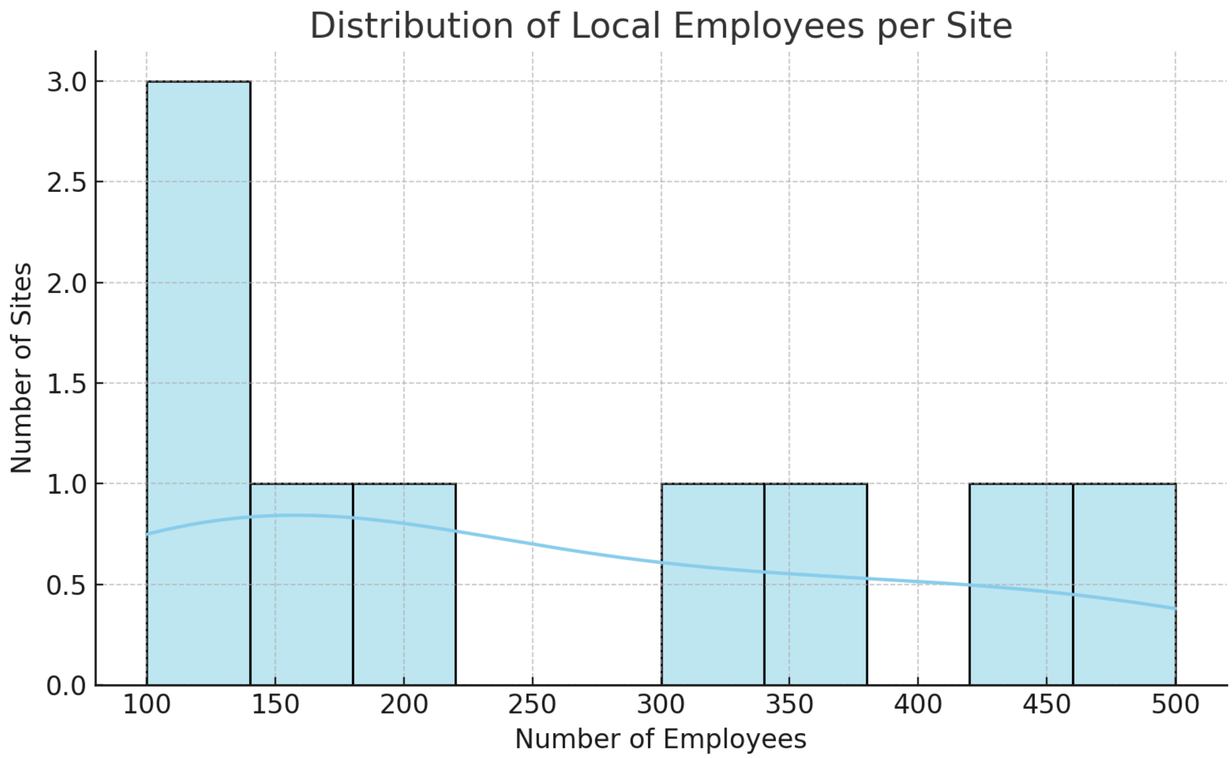
Mapping Disparities in Access to Auto Industry Jobs

To support our broader analysis of employment and transportation access in Detroit, a series of visualizations was created to explore patterns in the local automobile industry. Since Ford, GM, and FCA remain key employers in the region, understanding where their job sites are located and how accessible they are to Detroiters relying on public transit is central to evaluating spatial inequality. These plots build on the interactive map previously discussed and offer more detailed insights into job concentration, distribution, and physical distance from the city's core.



The first plot is a bar chart comparing total local employees by company. This simple yet effective visualization reveals that Ford employs the most people locally among the Detroit Three, followed by GM and FCA. This matters because the concentration of jobs within one company can affect how employment trends impact specific neighborhoods. If a company like Ford were to reduce operations or relocate facilities, it could disproportionately affect certain communities. This bar chart also helps reinforce the significance of these three companies within Detroit's job landscape, emphasizing why the auto industry is a critical piece of the city's economy and why equitable access to these jobs is important.

The second visualization, a histogram of local employee counts across job sites, shows the distribution of job sizes within the industry. Most job sites have a relatively low number of employees, while only a few locations employ large numbers. This pattern suggests that while the auto industry does provide many jobs, they are not equally concentrated in every plant or facility. The majority of sites are small, meaning that even if a site is located near a transit-reliant neighborhood, the number of jobs it offers may be limited. On the other hand, the largest job sites which are often located farther from the city core may be harder to reach for those without a car. This imbalance reinforces the structural challenge posed by modal mismatch and underlines the importance of geography when evaluating access to opportunity.



Together, these two visualizations deepen the story our project is telling. They connect the historical roots of racial segregation and redlining with the present-day layout of the city's job market. While Detroit is often known for its cars, the irony is that those who build and service the industry may struggle the most to reach the factories and offices where those jobs exist. This disconnect speaks directly to the broader theory of spatial mismatch and the ways in which transportation systems either bridge or reinforce inequality. As our group has shown through regression, transit analysis, and spatial mapping, car access remains one of the strongest predictors of employment in high-demand industries. These plots help visualize why and also remind us that unless these structural issues are addressed, Detroit's economy will continue to leave behind the very people it depends on most.

Connecting "Mapping Detroit" to Our Analysis of Transit and Job Access

The first section of Mapping Detroit: Land, Community, and Shaping a City underscores the idea that Detroit's spatial layout is not accidental. It is the result of political, economic, and racial decisions made over decades. The editors argue that maps of Detroit reveal far more than geography; they uncover layers of inequality, displacement, and community persistence. These maps, in many ways, are tools that reflect power by showing who gets resources, who is pushed out, and who is left behind. This connects directly to the focus of our project: how the spatial legacy of redlining and transportation policy continues to shape access to economic opportunity in Detroit.

In particular, Mapping Detroit shows how housing, land use, and infrastructure have worked together to divide the city across racial and class lines. The book highlights how public decisions about road placement, industrial zones, and housing developments disproportionately affected Black residents, often isolating them from new investments and employment centers. This context is vital when considering our own data visualizations, especially the scatter plot of job site distance from the urban core. Just as the book illustrates how infrastructure serves to disconnect rather than unite, our findings reveal that major job hubs in the auto industry are often located far from transit-reliant neighborhoods. These geographic decisions are not neutral.They reflect the structural inequalities that Mapping Detroit warns about.

The authors also stress the role of community memory and localized experience in shaping how space is used and contested. Our project mirrors this theme by showing how transit-dependent residents are forced to navigate not only long distances but also unreliable systems that fail to match their schedules and needs. Just as Mapping Detroit uses maps to tell stories of disinvestment and resilience, we use maps and data plots to visualize barriers to employment, showing how structural forces continue to limit opportunity even when jobs technically exist.

Together, both our project and Mapping Detroit ask the same critical question: Who gets access to the city, and on what terms? Whether through land use, housing, or transportation, the answer too often reflects historical injustices. By layering current transit and job access data over these patterns, our analysis continues the work this book begins.Mapping is not just Detroit's streets, but its systems of exclusion and the ongoing need for just, people-centered planning.

Health Access and Transportation: A Compounding Crisis

While much of this project focuses on transportation access and employment opportunity, the same geographic and racial divides created by redlining and disinvestment also contribute to stark health disparities across Detroit. Public health cannot be separated from the systems that shape daily life including transit, income, neighborhood conditions, and car ownership. These social determinants of health play a key role in who gets to live a longer, healthier life in the city.

CDC data from the USALEEP project shows a dramatic 18-year life expectancy gap between neighborhoods in Detroit. Residents of transit-poor areas such as Brightmoor and Osborn can expect to live only into their early 60s, while wealthier or better-connected areas like Midtown reach up to 80 years. These disparities correlate closely with transportation insecurity, low car ownership, poor access to healthcare facilities, and historical underinvestment. Health outcomes, in this way, are another consequence of the spatial mismatch created by redlining.

Access to care is also unevenly distributed. While major hospitals such as Detroit Medical Center (DMC) and Henry Ford Health are concentrated near Midtown and Downtown, peripheral neighborhoods like Springwells and Chandler Park are recognized by SEMCOG as healthcare deserts. Residents in these areas often live in Medically Underserved Areas (MUAs) and must navigate multiple transportation barriers to receive care, particularly those without a vehicle. According to SEMCOG, nearly 24% of Detroit households do not own a car, compared to just 7% regionally. Without reliable buses, basic health services become distant and difficult to reach.

These challenges are reinforced by data from the CDC PLACES Explorer and Health Opportunity Index (HOI), which identify neighborhoods like Dexter-Linwood and Warrendale as having the lowest health opportunity scores. These same areas report the highest rates of chronic stress, poor physical health, and inactivity. These are not only indicators of current health outcomes but also predictors of long-term generational wellbeing. The stress of unreliable transit, job insecurity, and disconnection from essential services translates into worse health metrics at nearly every level.

The geographic contrast between neighborhoods south of 8 Mile Road (Detroit proper) and those north of 8 Mile suburbs makes these patterns even clearer. Within the city, rates of chronic disease, poor mental health, and limited preventive care are significantly higher. Suburban communities, by contrast, enjoy better access to transportation, more consistent healthcare coverage, and stronger public health indicators.

In sum, transportation in Detroit is not just about getting to work. It is a direct line to health, stability, and opportunity. The same systems that separate people from good jobs also separate them from doctors, healthy food, and safe environments. Addressing Detroit's transportation challenges, then, is also a public health intervention that must be rooted in equity, history, and community need.

Bibliography

Gallagher, Rebecca, et al., editors. Mapping Detroit: Land, Community, and Shaping a City. Wayne State University Press, 2015.